	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 4

Lab Title: Boolean Expression Simplification and Implementation

Student Name: Muammad Waqas **Reg. No:** 240756

Objective: To understand Boolean algebra ,expression and K-MAP in logic circuits

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

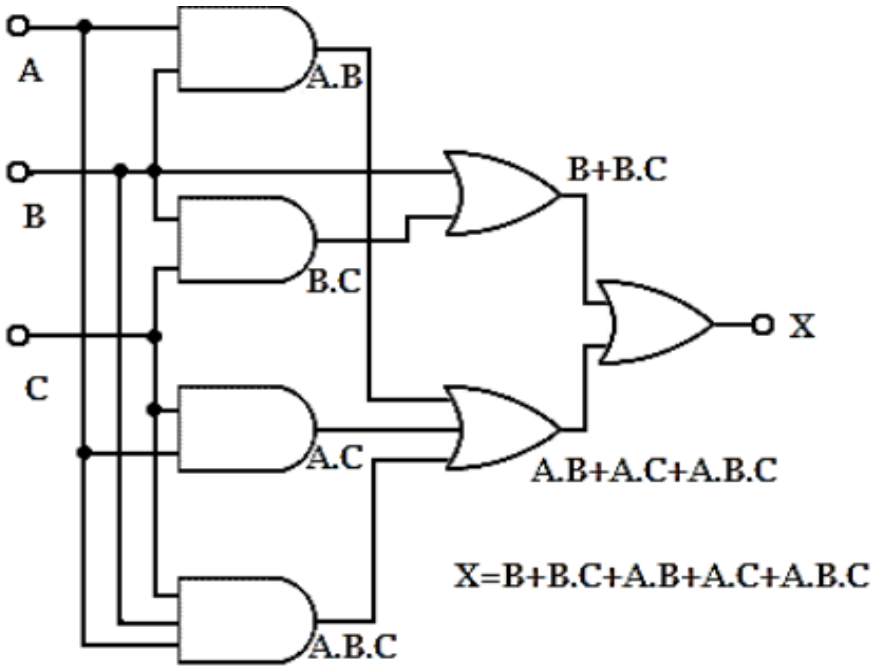
LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

TASK1:



Logic Expression:

$$X = B + B \cdot C + A \cdot B + A \cdot C + A \cdot B \cdot C$$

A	B	C	A.B	A.C	B.C	A.B.C	F
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	1
0	1	1	0	0	1	0	1
1	0	0	0	0	0	0	0
1	0	1	0	1	0	0	1
1	1	0	1	0	0	0	1
1	1	1	1	1	1	1	1

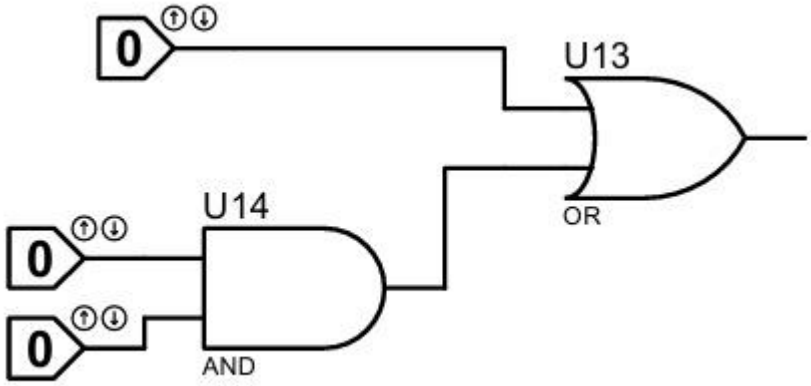
Simplification Of Expression:

X = B + B·C + A·B + A·C + A·B·C
= (B + B·C) + A·B + A·C + A·B·C
= B + A·B + A·C + A·B·C
= (B + A·B) + A·C + A·B·C
= B + A·C + A·B·C
= B + (A·C + A·B·C)
= B + A·C(1 + B)
= B + A·C
X = B + A·C

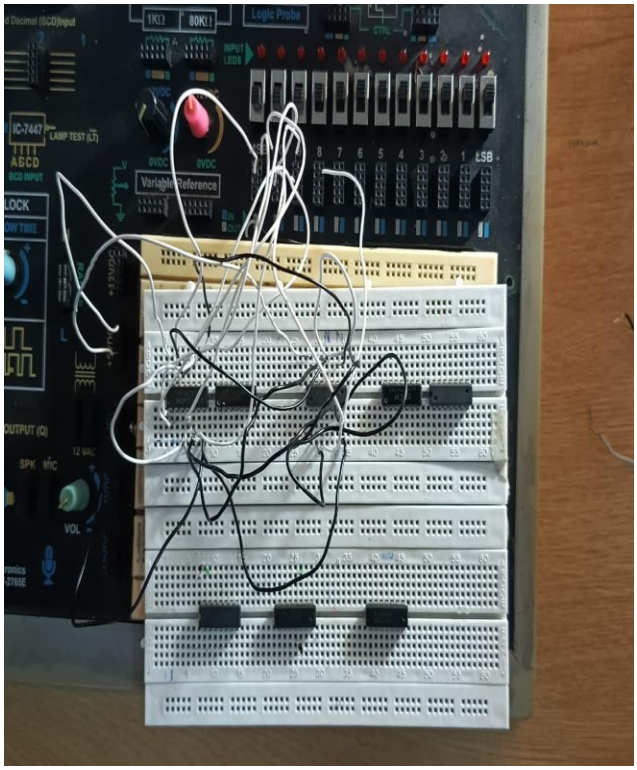
Truth table of simplified expression:

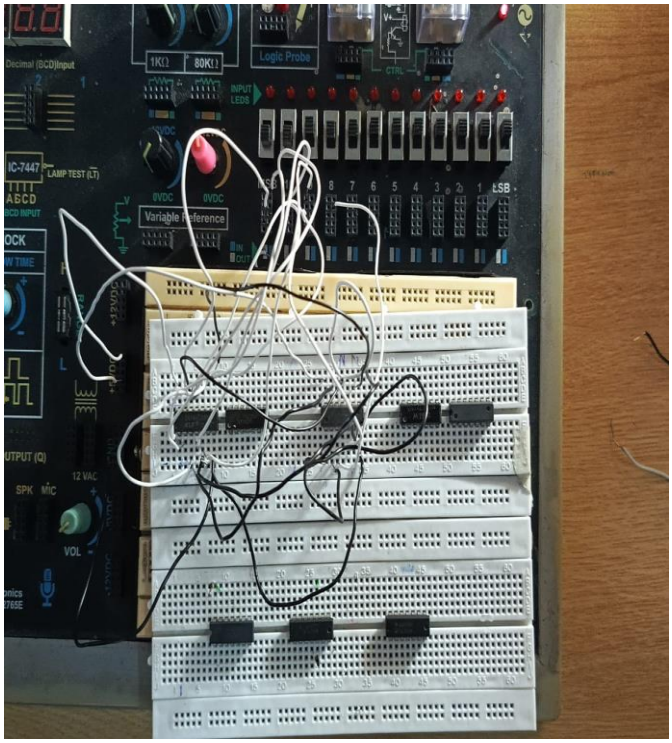
A	B	C	A.C	B+A.C	X
0	0	0	0	0	
0	0	1	0	0	
0	1	0	0	1	
0	1	1	0	1	
1	0	0	0	0	
1	0	1	1	1	
1	1	0	0	1	
1	1	1	1	1	

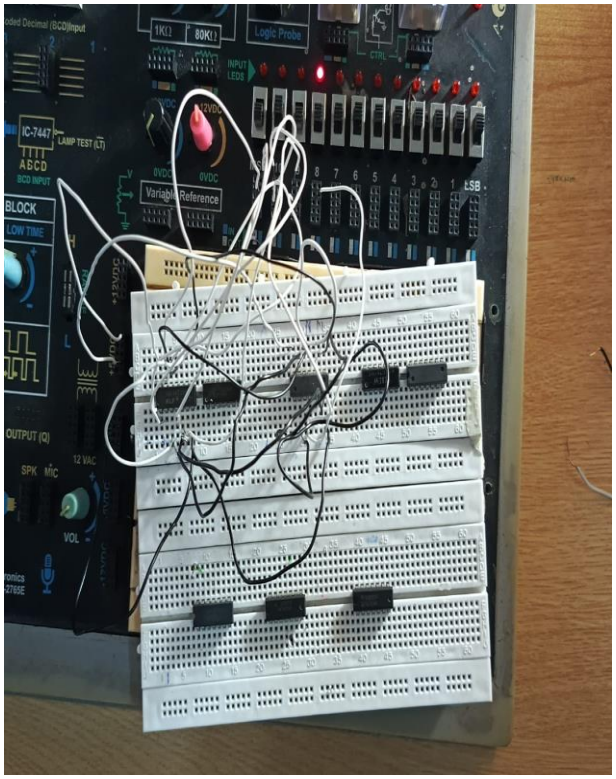
Modified Logic diagarm

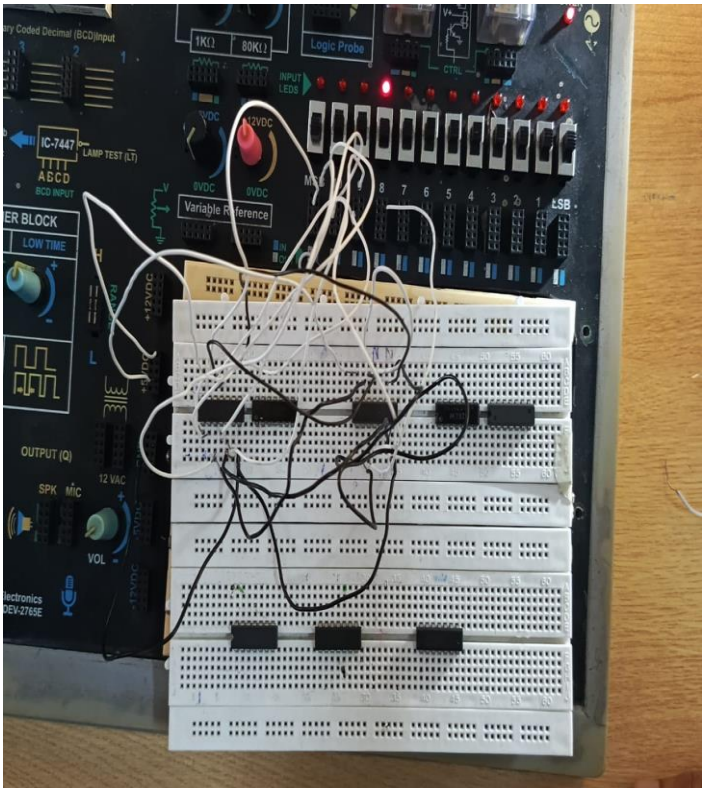


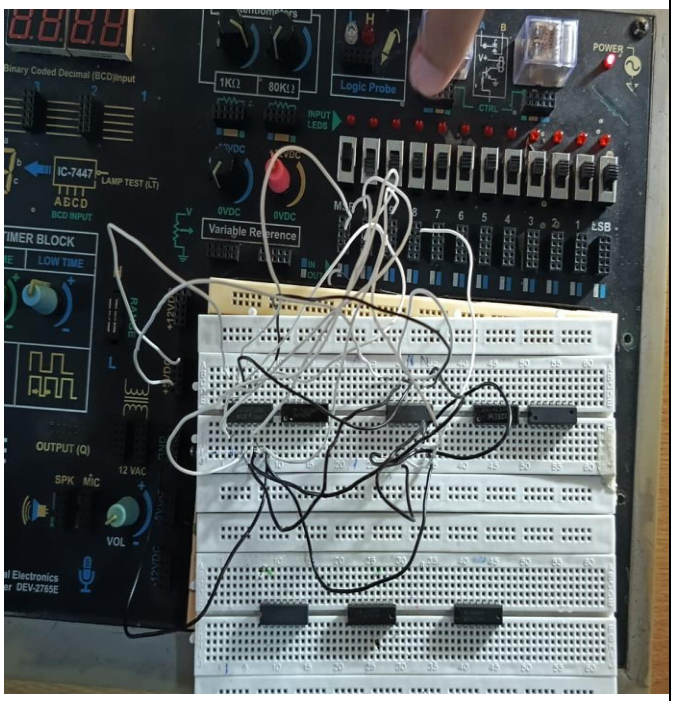
$$X = B + B \cdot C + A \cdot B + A \cdot C + A \cdot B \cdot C$$

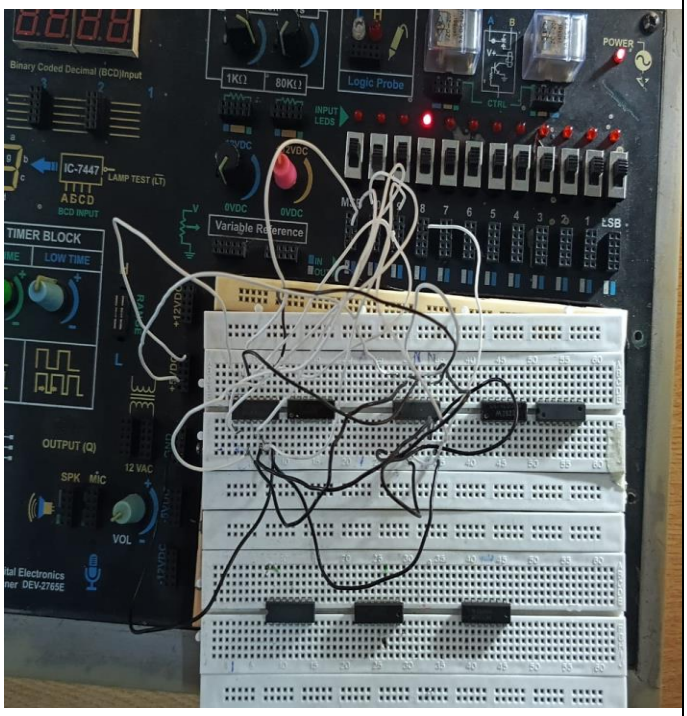


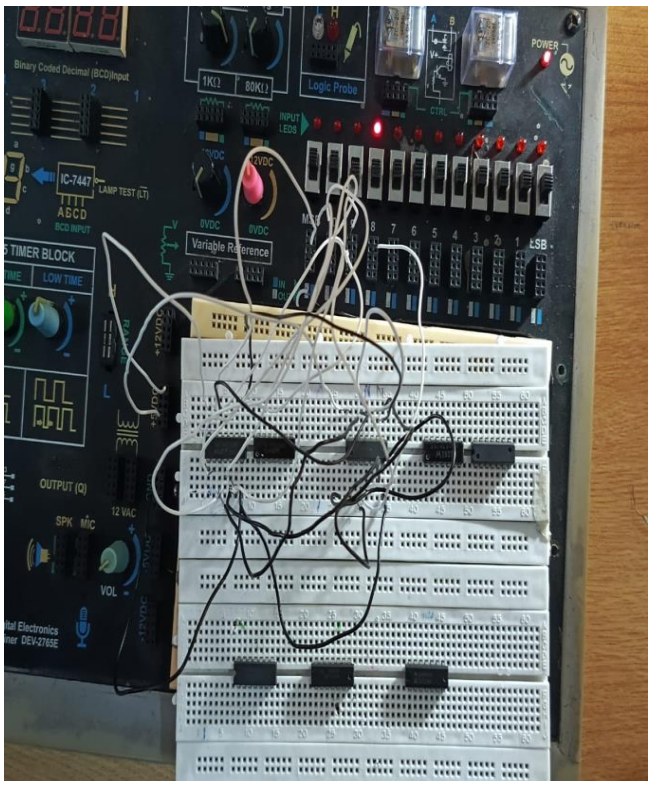


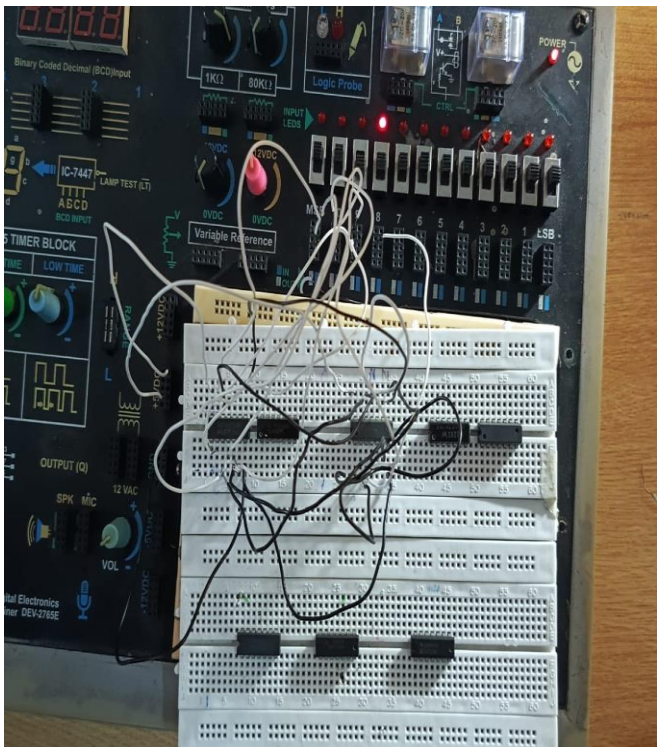




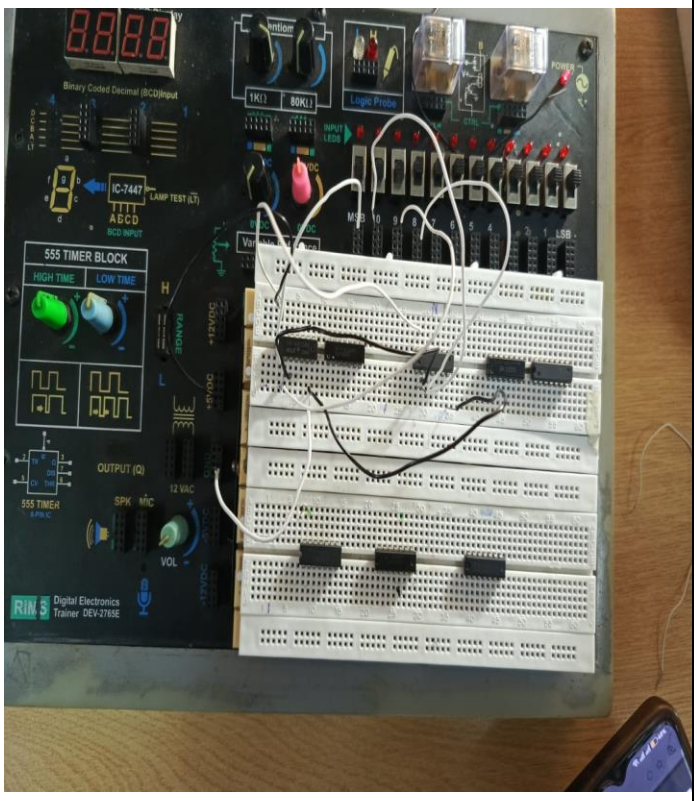

Binary Coded Decimal (BCD)Input
1KΩ
80KΩ
Logic Probe
POWER
IC-7447
LAMP TEST (LT)
ABCD
BCD INPUT
INPUT LEDS
0VDC
+2VDC
Variable Reference
LSB
TIMER BLOCK
LOW TIME
OUTPUT (Q)
12 VAC
SPK
MIC
VOL

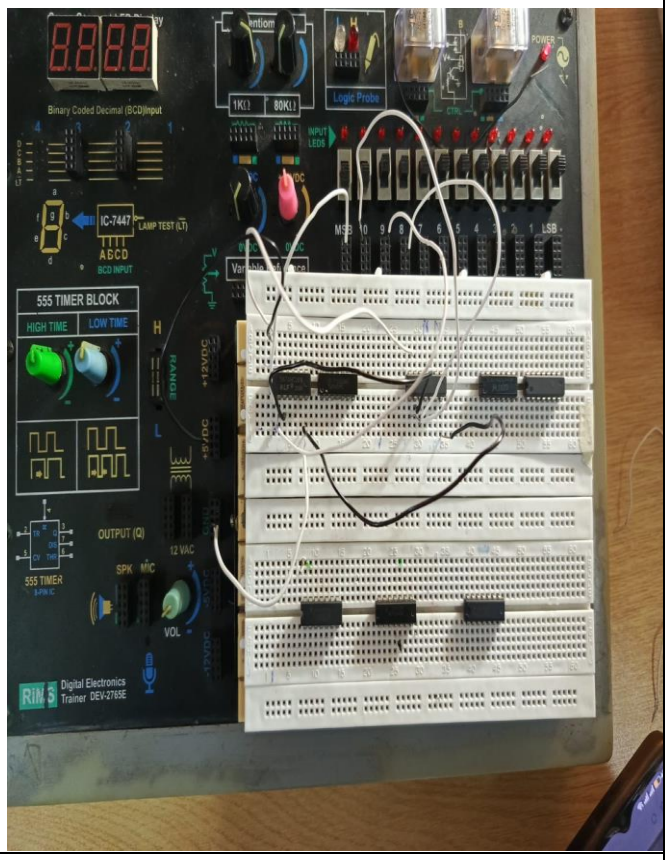

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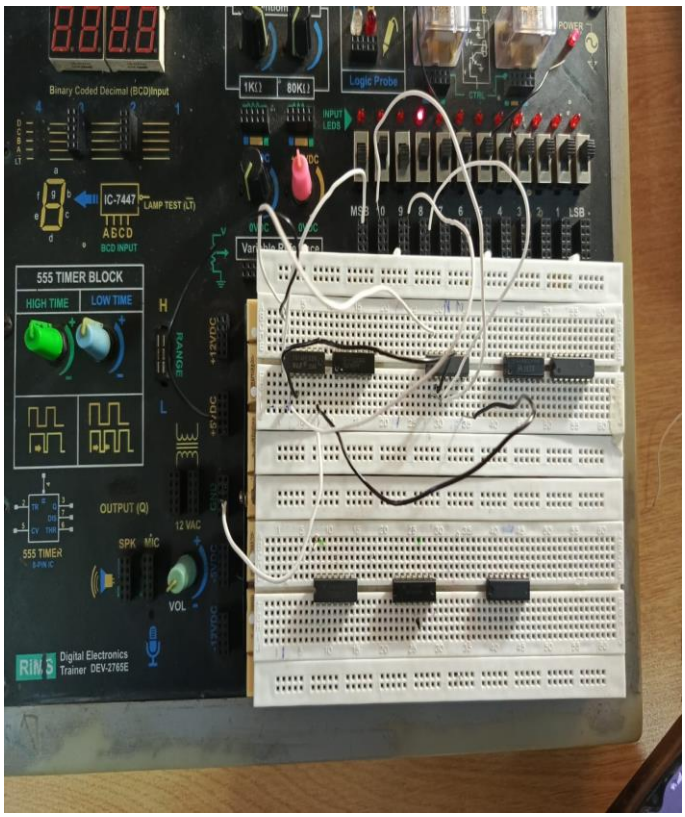

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X = B + A·C (Simplified Circuit Simulation)

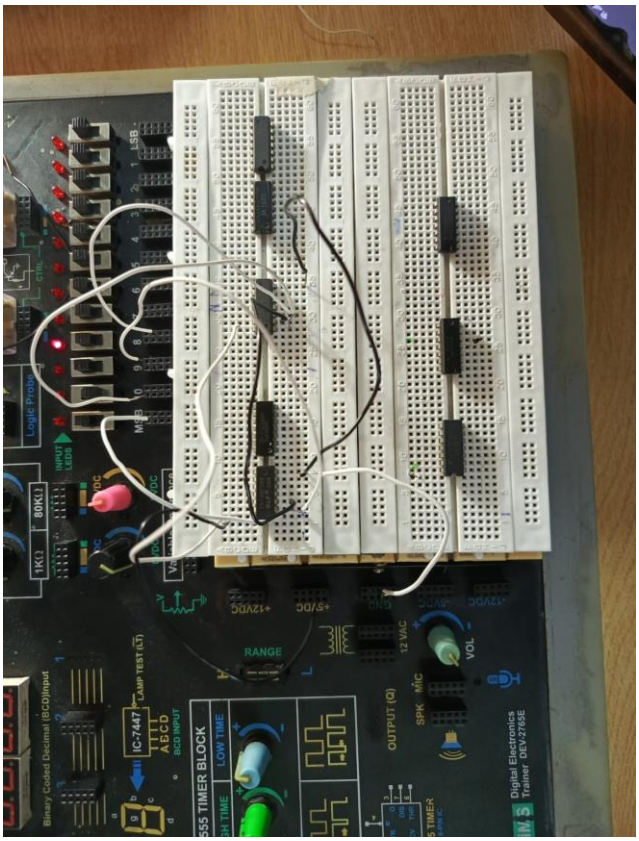


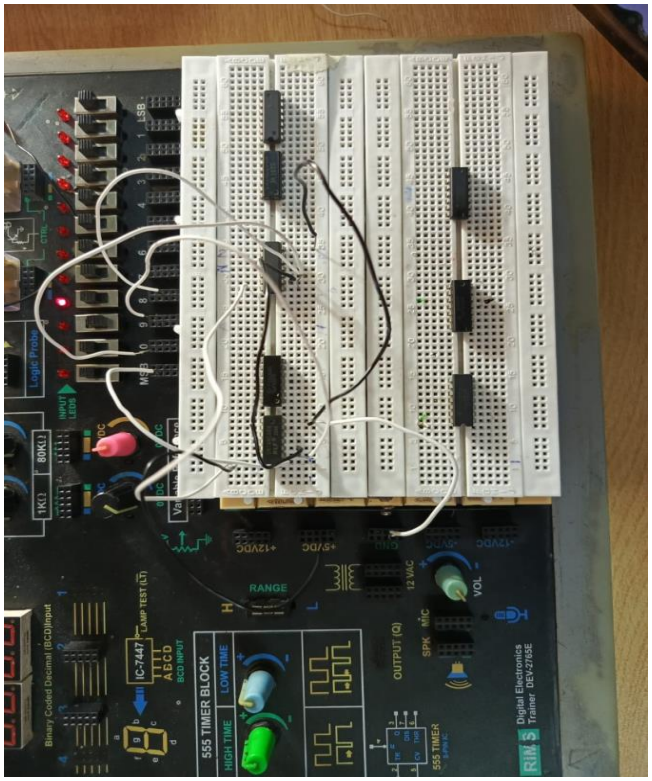






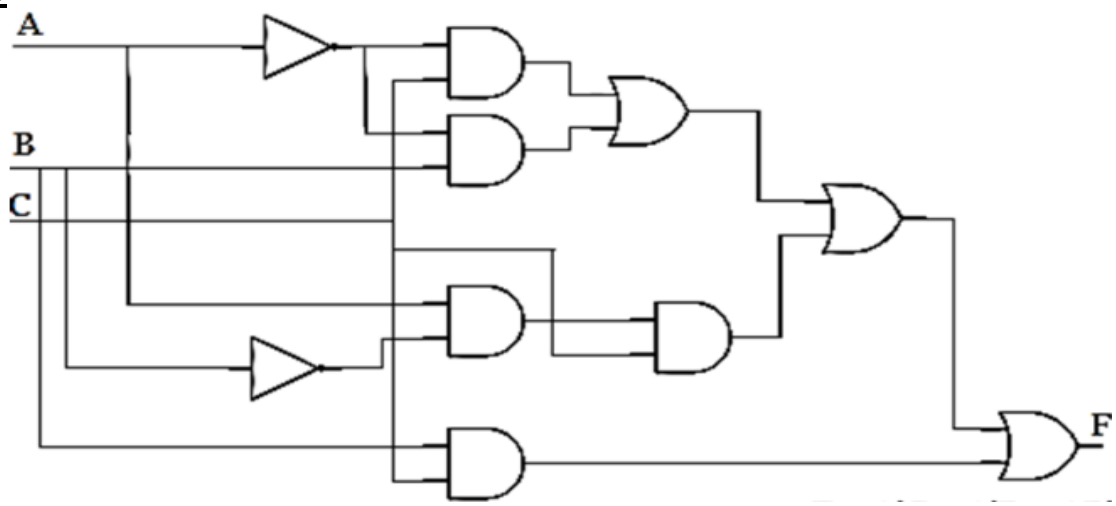

555 TIMER BLOCK
HIGH TIME
LOW TIME
IC-7447
LAMP TEST (LT)
BCD INPUT
OUTPUT (Q)
RANGE
VOL
Digital Electronics Trainer DEV-2765E


555 TIMER BLOCK
IC-7447
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BCD INPUT
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Digital Electronics Trainer DEV-2765E


555 TIMER BLOCK
IC-7447
OUTPUT (Q)
VOL
Digital Electronics Trainer DEV-2765E

Task 2:



Given Expression

F = A'C + A'B + AB'C + BC

1. Express in Sum of Minterms Form We will expand each term so that all variables (A, B, C) appear in every product. A'C = A'C(B + B') = A'B'C + A'BC A'B = A'B(C + C') = A'BC + A'BC' AB'C (already in 3 variables) BC = BC(A + A') = A'BC + ABC	**Now combine all minterms:** F = A'B'C + A'BC + A'BC + A'BC' + AB'C + A'BC + ABC **Remove duplicates:** F = A'BC' + A'B'C + A'BC + AB'C + ABC

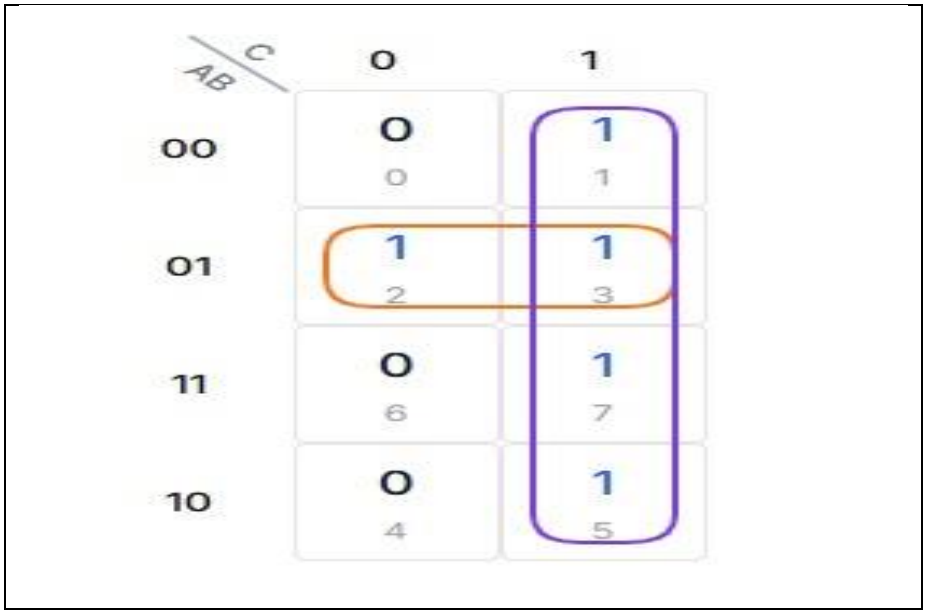
2: Write in Minterm (m) Notation

Now determine the binary for each term (A, B, C):

Term	A	B	C	Minterm
A'BC'	0	1	0	m2
A'B'C	0	0	1	m1
A'BC	0	1	1	m3
AB'C	1	0	1	m5
ABC	1	1	1	m7

F(A, B, C) = Σ(1, 2, 3, 5, 7)

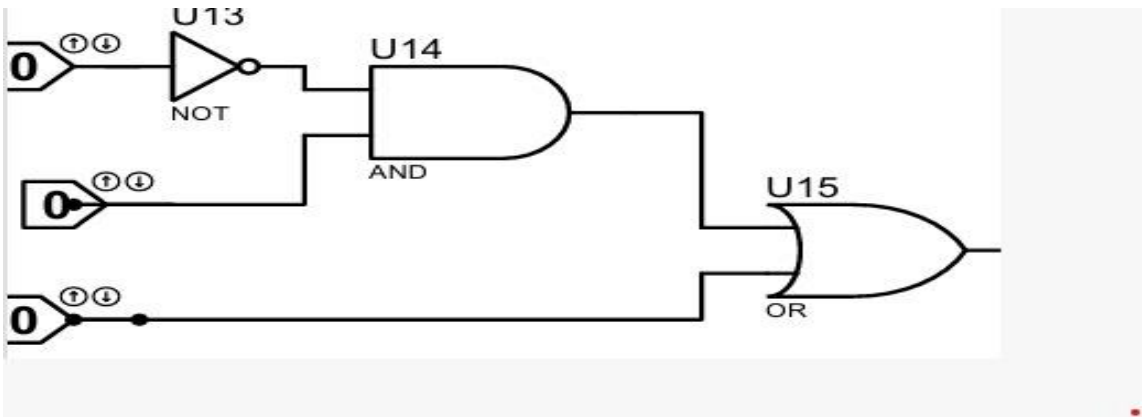
3 . K-MAP



Simplified Form

A*B+C

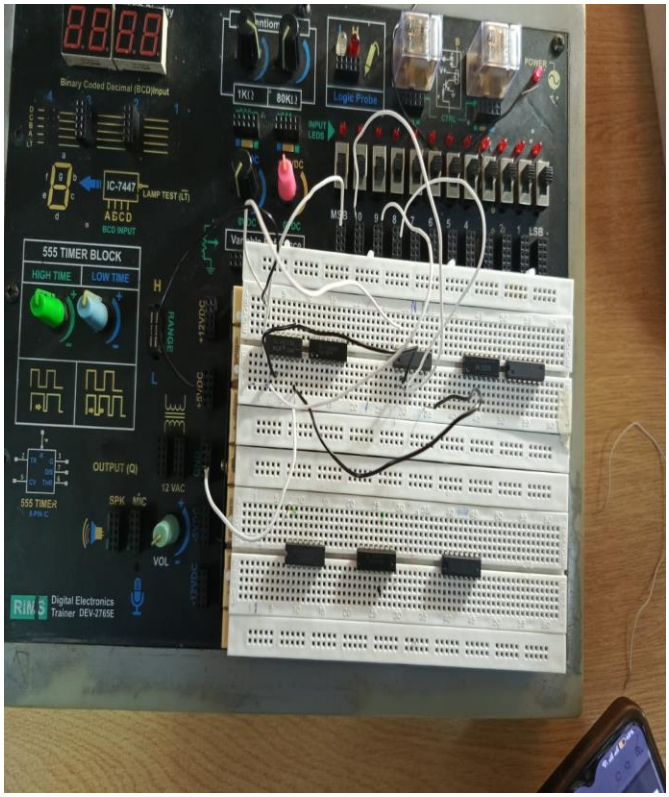
4. Simplified logic Diagram

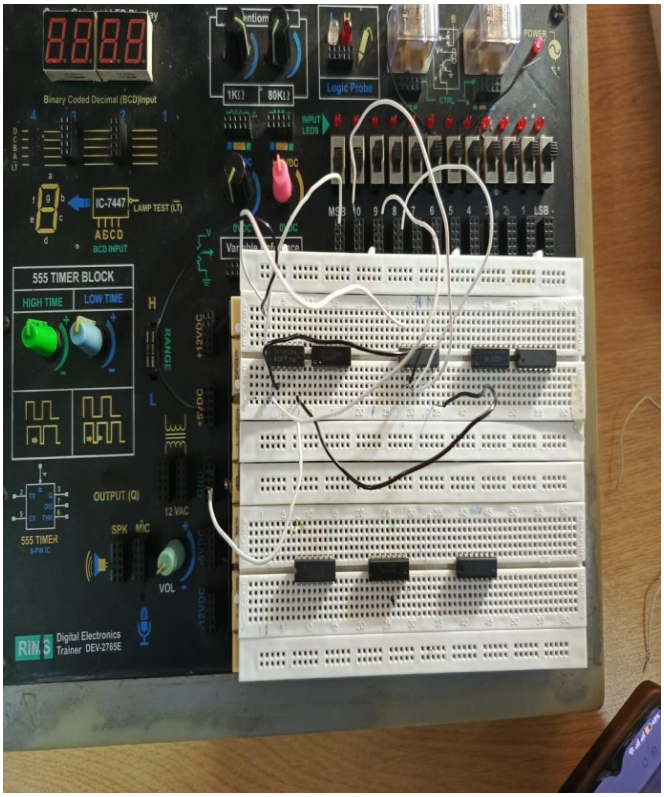


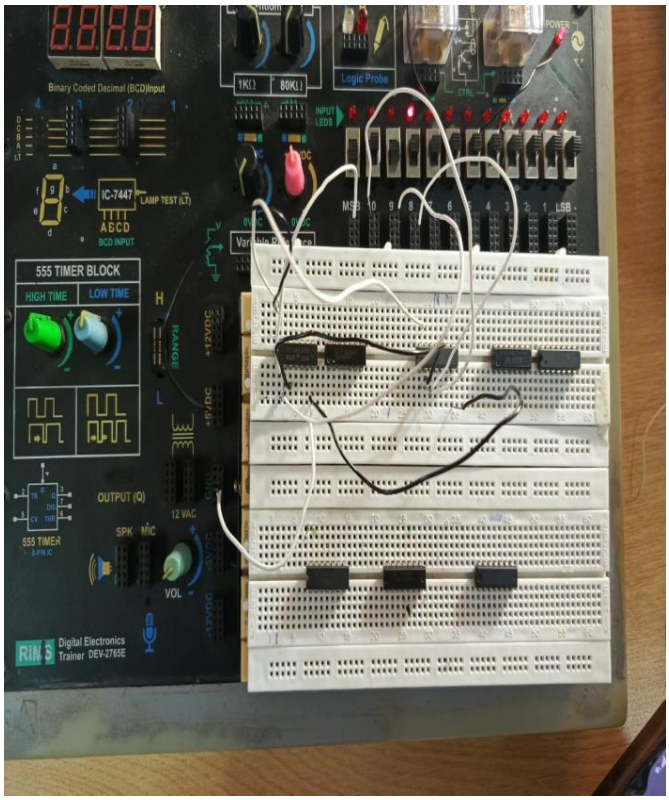
5. Truth Table for simplified expression:

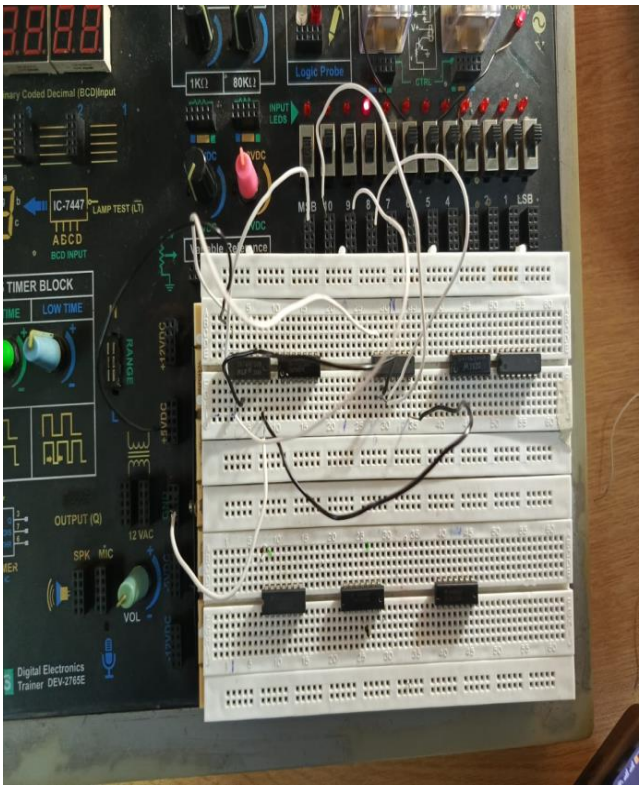
A	B	C	A*B	F
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	1
1	1	0	0	0
1	1	1	0	1

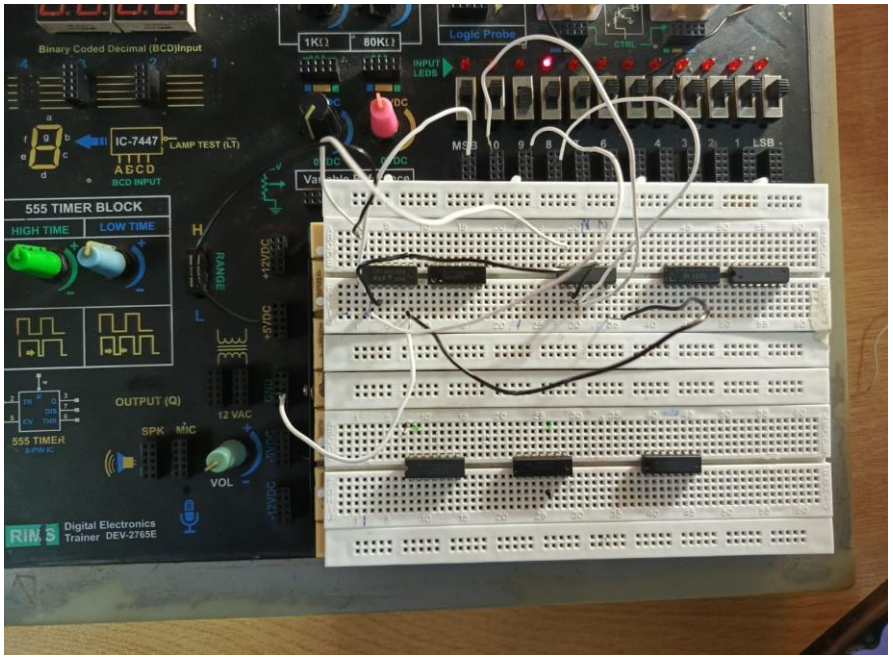
Circuit Simulation(Simplified Circuit)

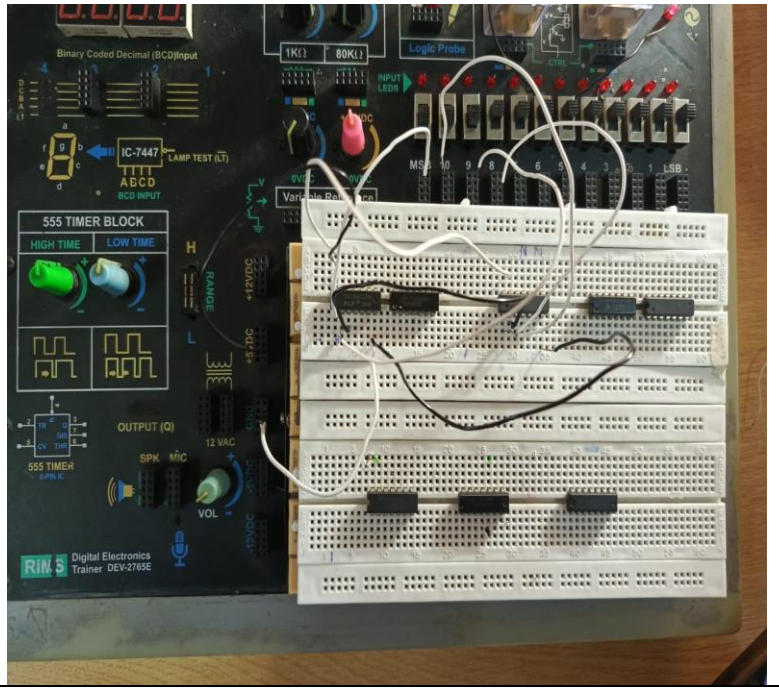


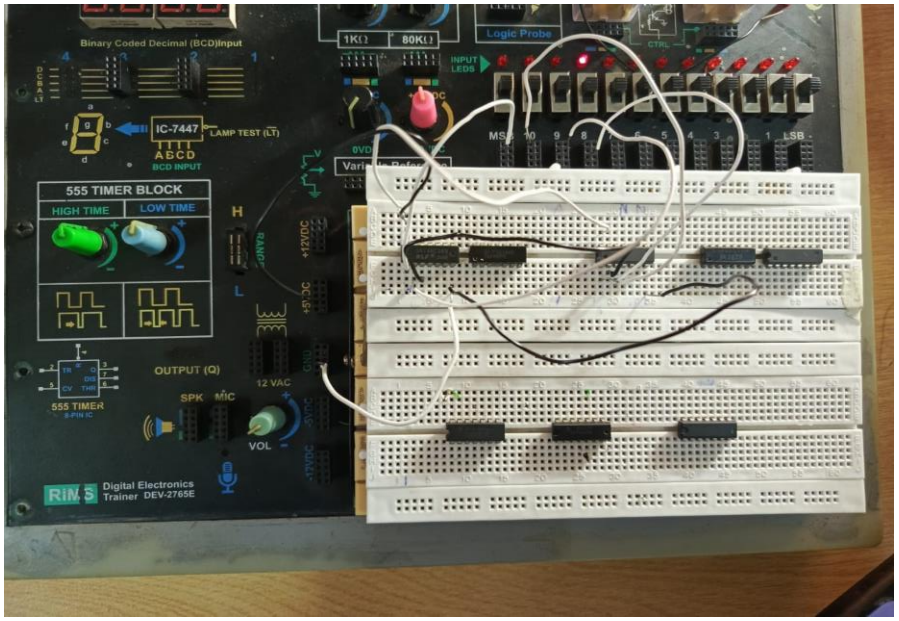


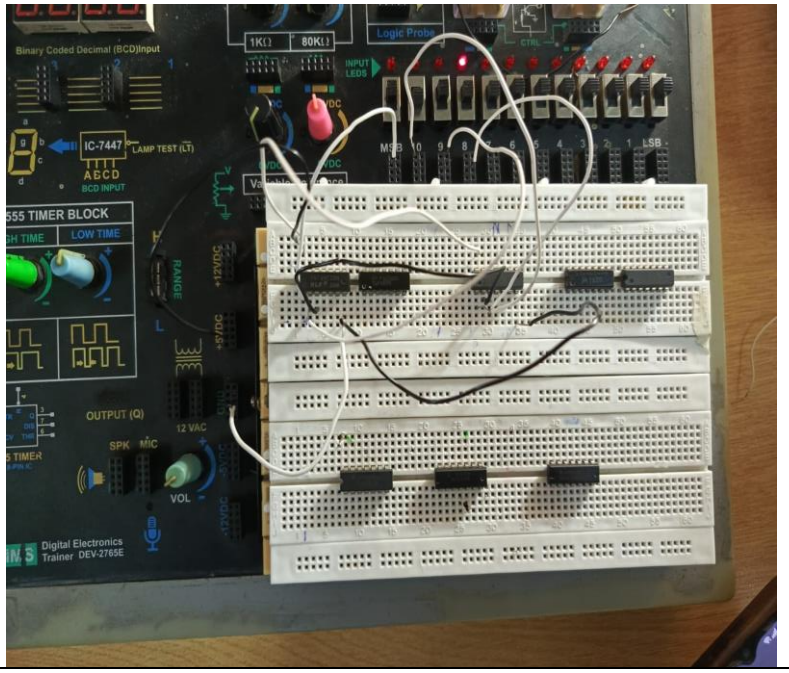





Binary Coded Decimal (BCD)Input
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LAMP TEST (LT)
A B C D
BCD INPUT
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HIGH TIME
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OUTPUT (Q)
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Digital Electronics
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Conclusion:

Boolean algebra helps us simplify complex logic expressions by applying simple rules like the distributive, associative, and absorption laws. This makes circuits easier to design and reduces the number of logic gates needed. A Karnaugh Map (K-map) is a visual tool that helps to simplify Boolean expressions by grouping together ones (1s) from the truth table. Using K-maps, we can quickly find the simplest Sum of Products (SOP) or Product of Sums (POS) form of a logic function. Overall, Boolean algebra and K-maps are powerful methods for designing efficient and cost-effective digital logic circuits.